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1. Introduction and Functions

The Radio Altimeter measures **height above ground level (AGL)** with high accuracy. It has three important functions:

1. **Flight deck display** of height AGL
2. **Automatic flight system data** — supplies height data for automatic landings when used with ILS/MLS
3. **GPWS** — furnishes height information and rate of change of height to the Ground Proximity Warning System

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2. FMCW Principle of Operation

The radio altimeter uses **Frequency Modulated Continuous Wave (FMCW)** transmission in an elliptical pattern vertically below the aircraft.

How it measures height:

1. Transmitter continuously sweeps frequency from start level **± 50 MHz, 300 times per second**
2. Radio wave travels to ground and back — during this travel time the transmitted frequency has changed
3. Equipment compares **transmitted vs. received frequency** — the difference = time of flight
4. From time of flight → height is calculated

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```
graph LR
  A["Tx frequency
continuously swept
±50 MHz / 300 Hz"] --> B["Wave travels to
ground and back"]
  B --> C["Rx frequency
(older, reflected signal)"]
  A --> D["Beat frequency
= Tx − Rx"]
  C --> D
  D --> E["= Time of flight
= Height AGL"]
```

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Handling the frequency reversal problem: When the transmitter reverses its sweep direction, a frequency breakdown occurs. This is overcome by averaging the **beat frequency over a short sampling period** — the changeover points are ignored.

3. Frequencies and Antenna

Band	Frequency Range	Status
SHF	**4200 MHz to 4400 MHz**	**Currently used**
UHF	1600 MHz to 1660 MHz	Previously used — obsolete

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Antenna characteristics:

- Flush-mounted horn antenna — transmits downward
- Conical/elliptical beam — wide enough to allow some portion to travel vertically even at pitch angles of **±30°** and roll angles of **±60°**
- Height determined from the **shortest path to ground** — always the vertical path
- Transmission is **continuous** → separate transmit and receive antennas are required (to avoid interference)
- Radiated power: **~1 watt**

4. Basic Indicator

Height Scale

The scale is **logarithmic**: expanded from **0–500 ft**, then non-linear from **500–2500 ft**.

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Mask

The height pointer disappears behind a mask when:

- Altitude exceeds **2500 ft**
- Any fault in the transmitted signal
- Altimeter is switched off

Failure Warning Flag

Appears when:

- Too much radio noise (corrupts returning signal)

- Local reflections received from airframe
- Loss of power to the equipment

Press-to-Test Button / Height Selector (dual purpose)

- **Pressed:** height pointer swings to a known pre-set altitude (confidence check)
- **Turned:** selects the reference Low Altitude Index (decision height marker)

Low Height Warning

Warning light illuminates if the aircraft flies **below the pre-selected height**. An alert tone sounds with increasing loudness from approximately **100 ft above** the decision height setting, then ceases suddenly at the DH.

5. EFIS Indicators

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Boeing Style — EFIS Display

Boeing digital/analogue display behaviour:

- Above 1000 ft: digital read-out only (white)
- Below 1000 ft: pictorial altimeter dial drawn alongside digital read-out; icon **shrinks in size** as height decreases
- As DH is approached: colour changes from **white** → **flashing amber**

Additionally, the Radio Altimeter **desensitises the autopilot and flight director response to the ILS glidepath** in the latter stages of an approach.

Airbus Style — EFIS Display

Airbus (A300 style):

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- Height indication at the **base of the EADI/PFD** attitude display
- Above DH: **green numbers**
- Below DH: **amber, slightly bigger numbers**
- Enhanced by a **voice warning system** giving aural height callouts at significant heights (as agreed by manufacturer and airline)

6. Range and Accuracy

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Radio Altimeter Performance:

- Range: **0 to 2500 ft AGL**
- Overall accuracy: **± 3% of indicated height OR ± 1 ft** — whichever is the greater
- Error contributors: Doppler shift, step error in digital counting circuits, height lag

7. Aircraft Installation Delay (Cable Length Compensation)

The radio altimeter must indicate **zero height AGL as the main wheels touch down**. In practice, the design target is **±1 ft accuracy**.

Two sources of installation offset that must be compensated:

1. **Residual Height** — height difference between the antennas on the fuselage and the bottom of the trailing main wheels on the approach to touchdown. Varies with aircraft weight and oleo compression.
2. **Cable run length** — distance between avionics bay and antenna on the underside of the fuselage. On a B747 this may be up to 100 ft of cable; on a small corporate jet as little as 6 ft. Compensation is applied × 2 (Tx to antenna AND Rx from antenna).

Why the RA may read negative on the ground: The equipment is calibrated to read zero when the main wheels first *touch* the runway (trailing wheel attitude). When parked level on the ground, the antenna is **below** its calibrated position relative to the landing attitude — so the RA may show a small negative altitude. The B747, with its multi-wheel trailing assembly, typically reads **−8 ft** on the ground.

8. Key Facts Summary

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Parameter	Value
Modulation type	FMCW (Frequency Modulated Continuous Wave)
Frequency (current)	4200–4400 MHz (SHF band)
Frequency sweep	±50 MHz, 300 sweeps/second
Transmission power	~1 watt
Operating range	0–2500 ft AGL
Accuracy	±3% or ±1 ft (whichever greater)
Beam coverage	±30° pitch, ±60° roll
Mask activation	Above 2500 ft, fault, or switched off
Scale type	Logarithmic (0–500 ft expanded, 500–2500 ft non-linear)
B747 on-ground reading	Typically −8 ft
Key system it supports	GPWS, Autoland (ILS/MLS), EFIS

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Important connection: The Radio Altimeter is a **major component of GPWS**. It provides the crucial height AGL and rate-of-change-of-height inputs that GPWS needs to generate terrain proximity warnings.

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